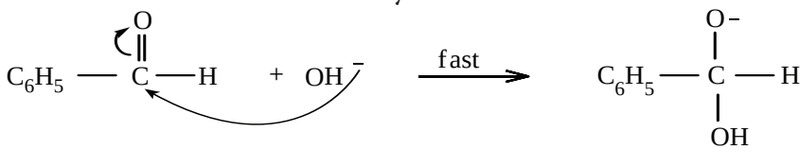
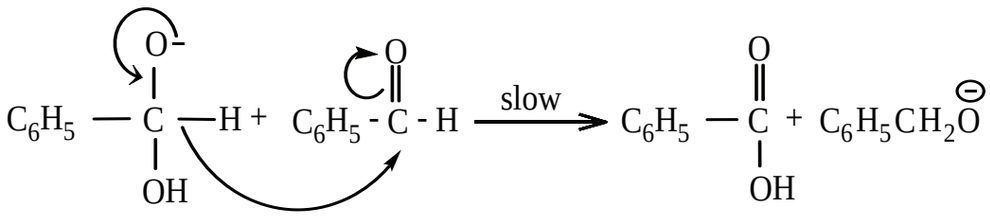
Mechanism of Cannizaro reaction

Cannizaro reaction involves three steps.

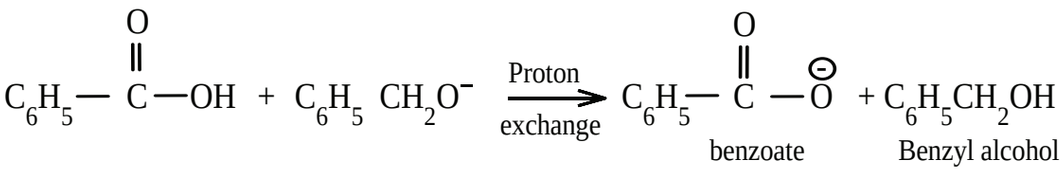
Step 1 : Attack of OH^- on the carbonyl carbon.



Step 2 : Hydride ion transfer



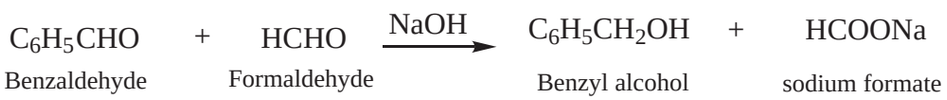
Step 3 : Acid – base reaction.



Cannizaro reaction is a characteristic reaction of aldehyde having no α – hydrogen.

Crossed Cannizaro reaction

When Cannizaro reaction takes place between two different aldehydes (neither containing an α hydrogen atom), the reaction is called as crossed cannizaro reaction.



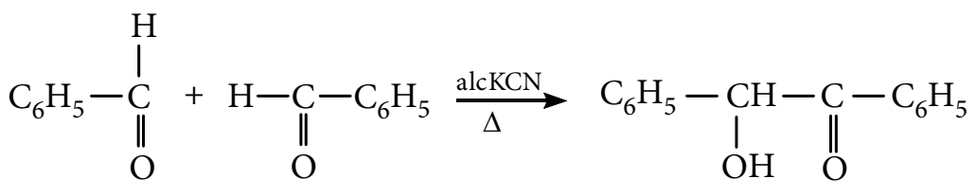
In crossed cannizaro reaction more reactive aldehyde is oxidized and less reactive aldehyde is reduced.

3) Benzoin condensation

The Benzoin condensation involves the treatment of an aromatic aldehyde with aqueous alcoholic KCN. The products are **α** hydroxy ketone.

Example

Benzaldehyde reacts with alcoholic KCN to form benzoin



Benzaldehyde

Benzoin

2-hydroxy – 1, 2 – diphenyl ethanone

XII U12-Carbonyl Compounds.indd 164 11-01-2021 16:10:39